Carias, R. D.
PID: A18573289

[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.

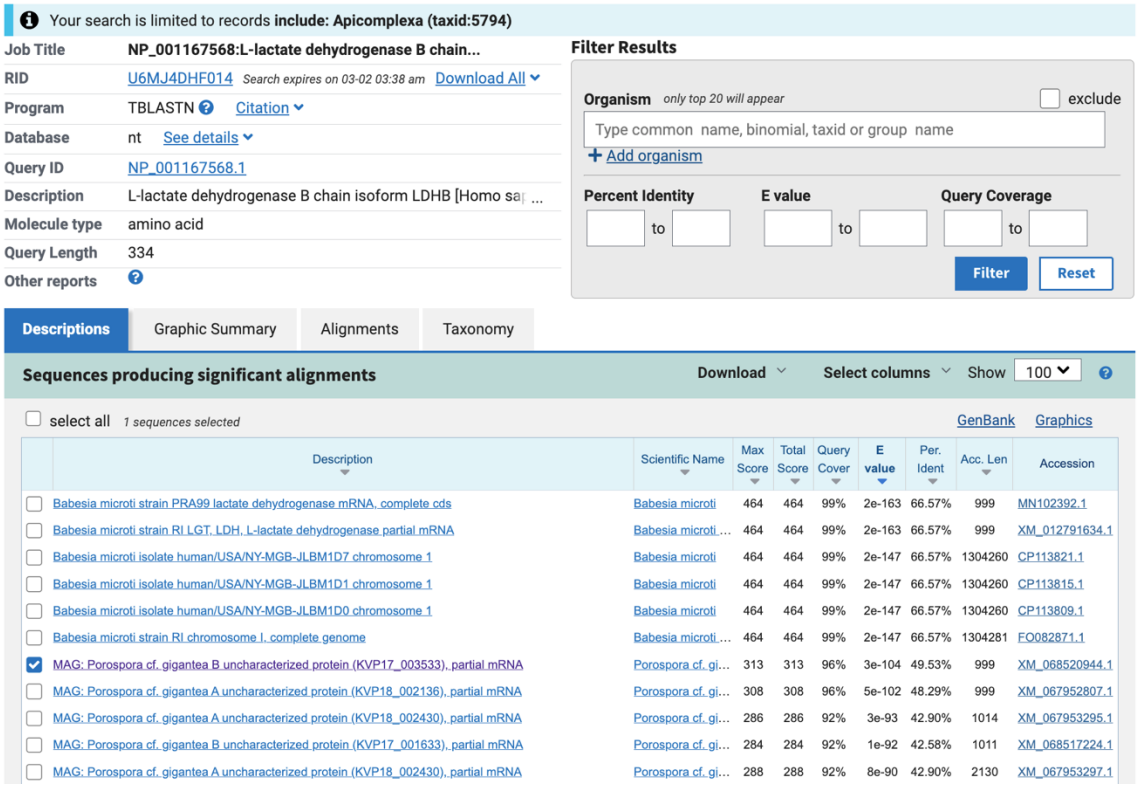
If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Name: Lactate Dehydrogenase B (LDHB)
Accession: **NP_001167568.1**
Species: Homo Sapiens

This gene encodes the B subunit of lactate dehydrogenase enzyme, which catalyzes the interconversion of pyruvate and lactate with concomitant interconversion of NADH and NAD+ in a post-glycolysis process. Alternatively spliced transcript variants have been found for this gene. Recent studies have shown that a C-terminally extended isoform is produced by use of an alternative in-frame translation termination codon via a stop codon readthrough mechanism, and that this isoform is localized in the peroxisomes. Mutations in this gene are associated with lactate dehydrogenase B deficiency. Pseudogenes have been identified on chromosomes X, 5 and 13. [provided by RefSeq, Feb 2016]

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN (2.7.1) search against Apicomplexan
Database: nt (DNA database)
Organism: Apicomplexa (Taxid: 5794)



Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Babesia microti strain PRA99 lactate dehydrogenase mRNA, complete cds	Babesia microti	464	464	99%	2e-163	66.57%	999	MN102392.1
Babesia microti strain RI LGT, LDH, L-lactate dehydrogenase partial mRNA	Babesia microti ...	464	464	99%	2e-163	66.57%	999	XM_012791634.1
Babesia microti isolate human/USA/NY-MGB-JLBM1D7 chromosome 1	Babesia microti	464	464	99%	2e-147	66.57%	1304260	CP113821.1
Babesia microti isolate human/USA/NY-MGB-JLBM1D1 chromosome 1	Babesia microti	464	464	99%	2e-147	66.57%	1304260	CP113815.1
Babesia microti isolate human/USA/NY-MGB-JLBM1D0 chromosome 1	Babesia microti	464	464	99%	2e-147	66.57%	1304260	CP113809.1
Babesia microti strain RI chromosome I, complete genome	Babesia microti ...	464	464	99%	2e-147	66.57%	1304281	FO082871.1
MAG: Porospora cf. gigantea B uncharacterized protein (KVP17_003533), partial mRNA	Porospora cf. gi...	313	313	96%	3e-104	49.53%	999	XM_068520944.1
MAG: Porospora cf. gigantea A uncharacterized protein (KVP18_002136), partial mRNA	Porospora cf. gi...	308	308	96%	5e-102	48.29%	999	XM_067952807.1
MAG: Porospora cf. gigantea A uncharacterized protein (KVP18_002430), partial mRNA	Porospora cf. gi...	286	286	92%	3e-93	42.90%	1014	XM_067953295.1
MAG: Porospora cf. gigantea B uncharacterized protein (KVP17_001633), partial mRNA	Porospora cf. gi...	284	284	92%	1e-92	42.58%	1011	XM_068517224.1
MAG: Porospora cf. gigantea A uncharacterized protein (KVP18_002430), partial mRNA	Porospora cf. gi...	288	288	92%	8e-90	42.90%	2130	XM_067953297.1

Chosen match:

Accession: **XM_068520944.1**, MAG: *Porospora cf. gigantea B* uncharacterized protein (KVP17_003533), partial mRNA.

Alignment statistics (Details below):

- Query coverage: 96%
- Percent identity: 49.53%
- E-value: 3e-104

Your search is limited to records **include: Apicomplexa (taxid:5794)**

Job Title	NP_001167568:L-lactate dehydrogenase B chain...
RID	U6MJ4DHF014 *Search expires on 03-02 03:38 am* Download All
Program	TBLASTN Citation
Database	nt See details
Query ID	NP_001167568.1
Description	L-lactate dehydrogenase B chain isoform LDHB [Homo sa ...
Molecule type	amino acid
Query Length	334
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Descriptions | Graphic Summary | **Alignments** | Taxonomy

Alignment view Pairwise Restore defaults Download

1 sequences selected

Download GenBank Graphics Next Previous Descriptions

MAG: Porospora cf. gigantea B uncharacterized protein (KVP17_003533), partial mRNA

Sequence ID: XM_068520944.1 **Length:** 999 **Number of Matches:** 1

Range 1: 55 to 993 GenBank Graphics Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
313 bits(802)	3e-104	Compositional matrix adjust.	159/321(50%)	213/321(66%)	10/321(3%)	+1

```
Query  14   EEEATVPNNKITVVGV-GQVGMACAISILGKSLADELALVDVLEDKLKGEMMDLQHGSLF  72
            +E T P   ITVVG  GQVG+ACA++ + +++ + + + DV+ D+L GE+ D+Q    F
Sbjct  55   RKEQTFPWT-ITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLHGEVKDIQQALTF  231

Query  73   LQTPKIV-ADKDYSVTANSKIVVVTAGVRQQEGESRLNLVQRNVNVFKFIIPQIVKYSPD  131
            +  P  V A  +   TANS+IVVVTAGVRQQ GESRL L+QRNVNVFK IIP+I K+SPD
Sbjct  232  IANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFKSIIPEIAKHSPD  411

Query  132  CIIIVVSNPVDILTYVTWKLSGLPKHRVIGSGCNLDSARFRYLMAEKLGIHPSSCHGWIL  191
             ++++ +NP DI+TYV WKLSG  ++RVI +GCNLD+ RFR+ +A +L     SS H +IL
Sbjct  412  AVLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANELHCSTSSIHAFIL  591

Query  192  GEHGDSSVAVWSGVNVAGVSLQELNPEMGTDNDSENWKEVHKMVVESAYEVIKLKGYTNW  251
            GEHGDSSV VWS  NV    + EL     T N        HK VV SA E+I+LKGYTNW
Sbjct  592  GEHGDSSVPVWSSTNVGSAIVSELCDLSKTSN-------AHKHVVASAEEIIRLKGYTNW  750

Query  252  AIGLSVADLIESMLKNLSRIHPVSTMVKGMYGIENEVFLSLPCILNARGLTSVINQKLKD  311
             IG SVA L   +L N   +  V T V+G++GIE +V L++PC+L A G+  V+N   KD
Sbjct  751  GIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGVHGVVNLTFKD  930

Query  312  DEVAQLKKSADTLWDIQKDLK  332
            +E   L+ SA TL +  K ++
Sbjct  931  EENEALRASAKTLAESMKGIE  993
```

Related Information

Gene - associated gene details

Genome Data Viewer - aligned genomic context

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[Q3] Gather information about this "novel" **protein**. At a minimum, show me the protein sequence of the "novel" protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don't forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don't have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

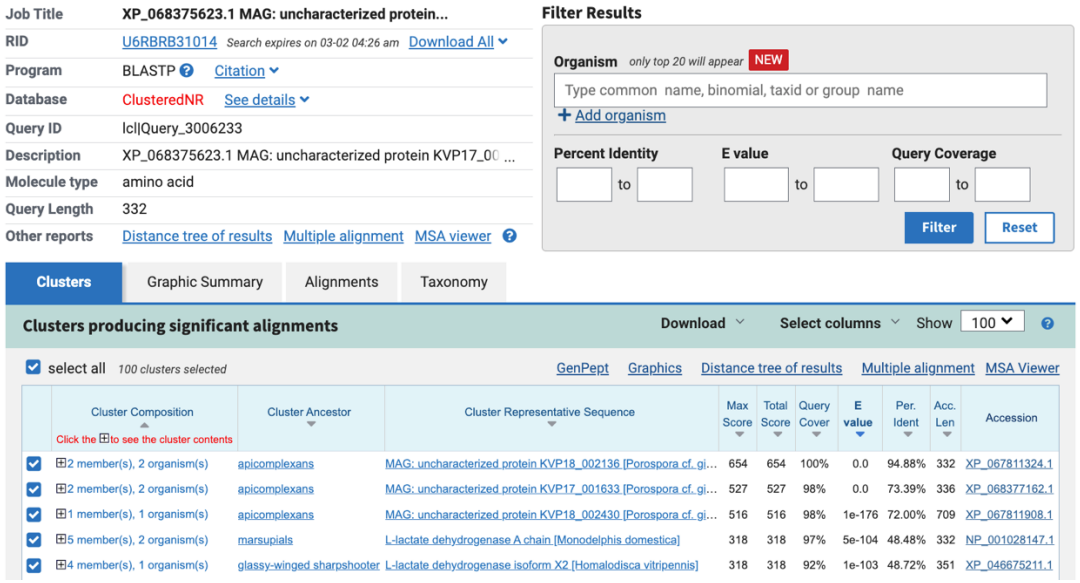
```
>XP_068375623.1 MAG: uncharacterized protein KVP17_003533 [Porospora cf. gigantea B]
MSAEDESSPVILNVFHTHRKEQTFPWTITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLHGE
VKDIQQALTFIANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFKSIIPEIAKHSPDA
VLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANELHCSTSSIHAFILGEHGDSSVPVW
SSTNVGSAIVSELCDLSKTSNAHKHVVASAEEIIRLKGYTNWGIGFSVAQLCRRILHNTEEVAAVGTFVQ
GLFGIEEDVVLTVPCLLGAHGVHGVVNLTFKDEENEALRASAKTLAESMKGIEY
```

Name: Uncharacterized protein (KVP17_003533)
Species: *Porospora cf. gigantea* B (MAG)

Eukaryota; Sar; Alveolata; Apicomplexa; Conoidasida; Gregarinasina; Eugregarinorida; Porosporidae; Porospora

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, "novel" is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

Details:
A BLASTP search against NR database (see setup in first screen-shot below) yielded a top hit result is to a protein from *Porospora cf. gigantea B* (MAG).
See additional screen shots below for top hits and selected alignment details:



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MAG: uncharacterized protein KVP18_002136 [Porospora cf. gigantea A]

Sequence ID: XP_067811324.1 Length: 332 Number of Matches: 1

Range 1: 1 to 332 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
654 bits(1687)	0.0	Compositional matrix adjust.	315/332(95%)	326/332(98%)	0/332(0%)

```
Query  1    MSAEDESSPVILNVFHTHRKEQTFPWTITVVGAAGQVGLACALACVSQAVGNRVCMFDVV  60
            M+ ED+SSPVILNVFHTHRKEQTFPWTIT++GAAGQVGLACALACVSQAVGNRVCMFDVV
Sbjct  1    MATEDDSSPVILNVFHTHRKEQTFPWTITIIGAAGQVGLACALACVSQAVGNRVCMFDVV  60

Query  61   ADRLHGEVKDIQQALTFIANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRN  120
            ADRL GEVKDIQQALTFIANPPCVFAAT+LK+TANSRIVVVTAGVRQQ GESRLGLLQRN
Sbjct  61   ADRLQGEVKDIQQALTFIANPPCVFAATDLKETANSRIVVVTAGVRQQSGESRLGLLQRN  120

Query  121  VNVFKSIIPEIAKHSPDAVLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHY  180
            VNVFKSIIPEIAK+SPDAVLLIATNPCDIM YVAWKLSGFCRNRVISTGCNLDTGRFRHY
Sbjct  121  VNVFKSIIPEIAKNSPDAVLLIATNPCDIMAYVAWKLSGFCRNRVISTGCNLDTGRFRHY  180

Query  181  LANELHCSTSSIHAFILGEHGDSSVPVWSSTNVGSAIVSELCDLSKTSNAHKHVVASAEE  240
            LANELHCSTSSIHAFILGEHGDSSVPVWSSTNVGSAIVSELCDLSKTSNAHKHVVASAEE
Sbjct  181  LANELHCSTSSIHAFILGEHGDSSVPVWSSTNVGSAIVSELCDLSKTSNAHKHVVASAEE  240

Query  241  IIRLKGYTNWGIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGV  300
            IIRLKG+TNWGIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGV
Sbjct  241  IIRLKGFTNWGIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGV  300

Query  301  HGVVNLTFKDEENEALRASAKTLAESMKGIEY  332
            HGVVNLTFK+EE++ L ASAKTLAESMKGIEY
Sbjct  301  HGVVNLTFKEEESDELLASAKTLAESMKGIEY  332
```

Related Information
Genome Data Viewer - aligned genomic context

Download GenPept Graphics ▼Next ▲Previous ⇥Descriptions

MAG: uncharacterized protein KVP17_001633 [Porospora cf. gigantea B]

Sequence ID: XP_068377162.1 Length: 336 Number of Matches: 1

Range 1: 10 to 336 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
527 bits(1357)	0.0	Compositional matrix adjust.	240/327(73%)	290/327(88%)	0/327(0%)

```
Query  6    ESSPVILNVFHTHRKEQTFPWTITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLH  65
            + SP+I N+F THRK+Q +PWTIT+VGAAGQVGLAC+LACV+Q VGNRV M+D  A++L
Sbjct  10   DHSPLIKNLFDTHRKDQRYPWTITIVGAAGQVGLACSLACVTQGVGNRVVMYDRSANKLQ  69

Query  66   GEVKDIQQALTFIANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFK  125
            GEV DI+QA +F++ PP V AA+ +++TANSR++V+TAGVRQQPGE+RL L+QRNV+++K
Sbjct  70   GEVNDIRQATSFMSVPPAVVAASTMEETANSRVIVITAGVRQQPGETRLSLVQRNVDIYK  129

Query  126  SIIPEIAKHSPDAVLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANEL  185
            SI+PE+AKHSP AVLLI TNPCDIM YVAWKLSGFCRNRVISTGCNLDTGRFRHYLA+EL
Sbjct  130  SIVPELAKHSPQAVLLIVTNPCDIMAYVAWKLSGFCRNRVISTGCNLDTGRFRHYLASEL  189

Query  186  HCSTSSIHAFILGEHGDSSVPVWSSTNVGSAIVSELCDLSKTSNAHKHVVASAEEIIRLK  245
            STSS+HAFILGEHGDSSVP+WSS +VGSA+V+++ D +  S+AH+ VV SAE+IIRLK
Sbjct  190  SVSTSSVHAFILGEHGDSSVPIWSSASVGSAMVNKIEDFANSSAHQRVVTSAEDIRLK  249

Query  246  GYTNWGIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGVHGVVN  305
            GYTNWGIGFSVA LCRRILH++EEVA++GTFVQGLFGIEEDVVL VPCLLG+HGVHGVVN
Sbjct  250  GYTNWGIGFSVANLCRRILHDSEEVASIGTFVQGLFGIEEDVVLAVPCLLGSHGVHGVVN  309

Query  306  LTFKDEENEALRASAKTLAESMKGIEY  332
            LT   EE E L+ASA+TLAE MK I++
Sbjct  310  LTMSKEEKEKLQASARTLAEIMKTIDF  336
```

Related Information
Genome Data Viewer - aligned genomic context

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MAG: uncharacterized protein KVP18_002430 [Porospora cf. gigantea A]

Sequence ID: XP_067811908.1 Length: 709 Number of Matches: 1

Range 1: 385 to 709 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
516 bits(1330)	1e-176	Compositional matrix adjust.	234/325(72%)	287/325(88%)	0/325(0%)

```
Query  8    SPVILNVFHTHRKEQTFPWTITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLHGE  67
            +P+I N+F THRK+Q +PWTIT+VGAAGQVGLAC+LACV+Q VGNRV M+D  AD+L GE
Sbjct  385  NPLINKLFDTHRKDQRYPWTITIVGAAGQVGLACSLACVTQGVGNRVMYDRSADKLRGE  444

Query  68   VKDIQQALTFIANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFKSI  127
              DI+QA +F++ PP V AA+ +++TANSR++V+TAGVRQQPGE+RL L+QRNV+++K++
Sbjct  445  CCDIRQATSFMSVPPAVVAASTIEETANSRVIVITAGVRQQPGETRLSLVQRNVDIYKTL  504

Query  128  IPEIAKHSPDAVLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANELHC  187
            IP +AKHSP AVLLI TNPCDIM YVAWKLSGFCRNRVISTGCNLDTGRFRHYLA+EL
Sbjct  505  IPALAKHSPQAVLLIVTNPCDIMAYVAWKLSGFCRNRVISTGCNLDTGRFRHYLASELSV  564

Query  188  STSSIHAFILGEHGDSSVPVWSSTNVGSAIVSELCDLSKTSNAHKHVVASAEEIIRLKGY  247
            STSS+HAFILGEHGDSSVP+WSS +VGSA+V+++ D +  S+AH+ VV SAE+IIRLKGY
Sbjct  565  STSSVHAFILGEHGDSSVPIWSSASVGSAMVNKIEDFANSSAHERVVTSAEDIIRLKGY  624

Query  248  TNWGIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGVHGVVNLT  307
            TNWGIGFSVA LCRRILH++EEVA++GTFVQGLFGI+EDVVL VPCLLG+HGVHGVVNLT
Sbjct  625  TNWGIGFSVANLCRRILHDSEEVASIGTFVQGLFGIKEDVVLAVPCLLGSHGVHGVVNLT  684

Query  308  FKDEENEALRASAKTLAESMKGIEY  332
              ++E   L+ASA+TLAE MK I++
Sbjct  685  MSEDEKNKLQASARTLAEIMKTIDF  709
```

Related Information
Genome Data Viewer - aligned genomic context

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L-lactate dehydrogenase A chain [Monodelphis domestica]

Sequence ID: NP_001028147.1 Length: 332 Number of Matches: 1

Range 1: 8 to 332 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
318 bits(815)	5e-104	Compositional matrix adjust.	160/330(48%)	218/330(66%)	12/330(3%)

```
Query  10   VILNVFHTHRKEQTFPWTITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLHGEVK  69
            +ILNV    ++EQT    ITVVG  G VG+ACA++ + + + + + + DV+ D+L  GE+
Sbjct  8    LILNVL---KEEQTPHNKITVVGV-GAVGMACAISILMKDLADELALVDVIEDKLKGEMM  63

Query  70   DIQQALTFIANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFKSIIP  129
            D+Q   F+  P  V ++ +   TANS++VV+TAG RQQ GESRL L+QRNVN+FK IIP
Sbjct  64   DLQHGSLFLKTPKIV-SSKDYAVTANSKLVVITAGARQQEGESRLNLVQRNVNIFKFIIP  122

Query  130  EIAKHSPDAVLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANELHCST  189
            I K+SP+  LL+ +NP DI+TYVAWKLSGF +NRVI +GCNLD+ RFR+ +  +L    +
Sbjct  123  NIVKYSPNCKLLVSNPVDILTYVAWKLSGFPKNRVIGSGCNLDSARFRYLMGEKLGIHS  182

Query  190  SSIHAFILGEHGDSSVPVWSSTNVGSAIVSEL--------CDLSKTSNAHKHVVASAEEII  242
            SS H +ILGEHGDSSVPVWS  NV    +  L        D  +  + HK VV SA E+I
Sbjct  183  SSCHGWILGEHGDSSVPVWSGVNVAGVSLKSLHPALGTDSDSEQWKDVHKQVVESAYEVI  242

Query  243  RLKGYTNWGIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGVHG  302
            +LKGYT+W IG SVA L   I+ N    V  + T ++GL+GI EDV L+VPC+LG +G+
Sbjct  243  KLKGYTSWAIGLSVADLAESIMKNLRRVHPISTMIKGLYGINEDVFLSVPCILGQNGISD  302

Query  303  VVNLTFKDEENEALRASAKTLAESMKGIEY  332
            VV +T   EE   L+ SA TL    K +++
Sbjct  303  VVKVTLTTEEESRLKQSADTLWGIQKELQF  332
```

Related Information
Gene - associated gene details
AlphaFold Structure - 3D structure displays
Genome Data Viewer - aligned genomic context

Download GenPept Graphics ▼Next ▲Previous ⇥Descriptions

L-lactate dehydrogenase isoform X2 [Homalodisca vitripennis]

Sequence ID: XP_046675211.1 Length: 351 Number of Matches: 1

Range 1: 42 to 351 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
318 bits(814)	1e-103	Compositional matrix adjust.	152/312(49%)	217/312(69%)	9/312(2%)

```
Query  28   ITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLHGEVKDIQQALTFIANPPCVFAA  87
            ITVVG  GQVG+ACA + ++Q V + + + D V D+L  GE+ D+Q    F+ +P  +++
Sbjct  42   ITVVGV-GQVGMACAFSILTQYVTDEIALVDCVEDKLRGEMMDLQHGACFMRSPK-IYSG  99

Query  88   TNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFKSIIPEIAKHSPDAVLLIATNPC  147
            T+   TANS+I +V+AGVRQ+ GESRL L+QRN+++FK IIP++ KHSPD +LLI +NP
Sbjct  100  TDYGITANSKICIVSAGVRQKEGESRLNLVQRNLDIFKHIIPQLVKHSPDTILLIVSNPV  159

Query  148  DIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANELHCSTSSIHAFILGEHGDSSVPV  207
            D++++YVAWKLSG  +NRVI +G NLDT R R  ++ +L+ S  S+H +I+GEHGD+SVPV
Sbjct  160  DVLSYVAWKLSGLPKNRVIGSGTNLDTSRLRFLVSQKLNLSVESVHGYIIGEHGDTSVPV  219

Query  208  WSSTNVGSAIVSELCDLSKTSNA--------HKHVVASAEEIIRLKGYTNWGIGFSVAQLC  260
            WS  NV    + ++ D     N         HK VV SA EIIRLKGYT+W IG  VA+LC
Sbjct  220  WSGVNVAGVRLRDVNDDIGRKNDTESFNLIHKQVVDSAYEIIRLKGYTSWAIGLCVAKLC  279

Query  261  RRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGVHGVVNLTFKDEENEALRASA  320
            + ILHN   V AV     +G  GI++DV L++PC+LG +GV  V+     ++EE + L+ SA
Sbjct  280  QSILHNAHSVHAVSNAAKGFHGIDQDVFLSLPCVLGENGVSHVIKQPLREEELQQLKKSA  339

Query  321  KTLAESMKGIEY  332
            KT+A+ +KG+++
Sbjct  340  KTMADVIKGLKF  351
```

Related Information
Genome Data Viewer - aligned genomic context

Carias, R. D.
PID: A18573289

[Q5] Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequences.

>NP_001167568.1 L-lactate dehydrogenase B chain isoform LDHB [Homo sapiens] MATLKEKLIAPVAEEEATVPNNKITVVGVGQVGMACAISILGKSLADELALVDVLEDKLKGEMMDLQH GSLFLQTPKIVADKDYSVTANSKIVVVTAGVRQQEGESRLNLVQRNVNVFKFIIPQIVKYSPDCIIIVVSNP VDILTYVTWKLSGLPKHRVIGSGCNLDSARFRYLMAEKLGIHPSSCHGWILGEHGDSSVAVWSGVNVA GVSLQELNPEMGTDNDSENWKEVHKMVVESAYEVIKLKGYTNWAIGLSVADLIESMLKNLSRIHPVST MVKGMYGIENEVFLSLPCILNARGLTSVINQKLKDDEVAQLKKSADTLWDIQKDLKDL
>XP_068375623.1 MAG: uncharacterized protein KVP17_003533 [Porospora cf. gigantea B] MSAEDESSPVILNVFHTHRKEQTFPWTITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLHGE VKDIQQALTFIANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFKSIIPEIAKHSPDA VLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANELHCSTSSIHAFILGEHGDSSVPVW SSTNVGSAIVSELCDLSKTSNAHKHVVASAEEIIRLKGYTNWGIGFSVAQLCRRILHNTEEVAAVGTFVQ GLFGIEEDVVLTVPCLLGAHGVHGVVNLTFKDEENEALRASAKTLAESMKGIEY
>NP_001028147.1 L-lactate dehydrogenase A chain [Monodelphis domestica] MGTVKDQLILNVLKEEQTPHNKITVVGVGAVGMACAISILMKDLADELALVDVIEDKLKGEMMDLQH GSLFLKTPKIVSSKDYAVTANSKLVVITAGARQQEGESRLNLVQRNVNIFKFIIPNIVKYSPNCKLLVVSN PVDILTYVAWKLSGFPKNRVIGSGCNLDSARFRYLMGEKLGIHSSSCHGWILGEHGDSSVPVWSGVNVA GVSLKSLHPALGTDSDSEQWKDVHKQVVESAYEVIKLKGYTSWAIGLSVADLAESIMKNLRRVHPISTM IKGLYGINEDVFLSVPCILGQNGISDVVKVTLTTEEESRLKQSADTLWGIQKELQF
>NP_001080702.1 lactate dehydrogenase A S homeolog [Xenopus laevis] MATVKDKLIHNVVKEESLPQNKVTIVGVGAVGMACAISVLQKDLADELALVDVIEDKLKGEMMDLQH GSLFLRTPKIVSGKDYSVTANSKLVVVTAGARQQEGESRLNLVQRNVNIFKFIIPNIVKYSPNCTLLIVSNP VDILTYVAWKISGFPKNRVIGSGCNLDSARFRYLMGQKFGIHTQSCHGWVIGEHGDSSVPVWSGVNVA GVSLKTLHPDIGSDADKENWKEVHKQVVDSAYEVIKLKGYTSWAIGLSVADLSESILKNLRRVHPISTM VKGMYGVNNDVFLSVPCVLGNLGITDVVNMTLKADEEDRLRKSADTLWAIQKELQF
>NP_001261474.1 lactate dehydrogenase, isoform B [Drosophila melanogaster] MAAIKDSLLAQVAEVLPSSGHKVTIVGIGQVGMASAFSILAQNVSKEVCLIDVCADKLQGELMDLQHGS NFLKNPQITASTDFAASANSRLCIVTAGVRQKEGESRLSLVQRNTDILKNIIPKLVEYSPDTILLMVSNPV DIMTYVAWKLSGLPKNRVIGSGTNLDSSRFRFLMSQRLGVAPTSCHGWIIGEHGDSSVPVWSGVNIAGV RLRELNPILGTGEDPEKWNELHKQVVDSAYEVIKLKGYTSWAIGLSTASLASAILRNTSSVAAVSTSVLG EHGIDKDVFLSLPCVLNANGVTSVVKQILTPTEVEQLQKSANIMSDVQAGLKF
>XP_013843801.1 L-lactate dehydrogenase B chain isoform X1 [Sus scrofa] MRGGPRFLLLPCRSCARADARCRLCLLRATLTAVSRTFTPLRTMATLKEKLIAPVAEEETTIPNNKITVV GVGQVGMACAISILGKSLTDELALVDVLEDKLKGEMMDLQHGSLFLQTPKIVADKDYSVTANSKIVVV TAGVRQQEGESRLNLVQRNVNVFKFIIPQIVKYSPDCIIIVVSNPVDILTYVTWKLSGLPKHRVIGSGCNL DSARFRYLMAEKLGVHPSSCHGWILGEHGDSSVAVWSGVNVAGVSLQELNPEMGTDNDSENWKEVHK MVVESAYEVIKLKGYTNWAIGLSVADLIESMLKNLSRIHPVSTMVQGMYGIENEVFLSLPCVLNARGLT SVINQKLKDDEVAQLKNSADTLWGIQKDLKDL

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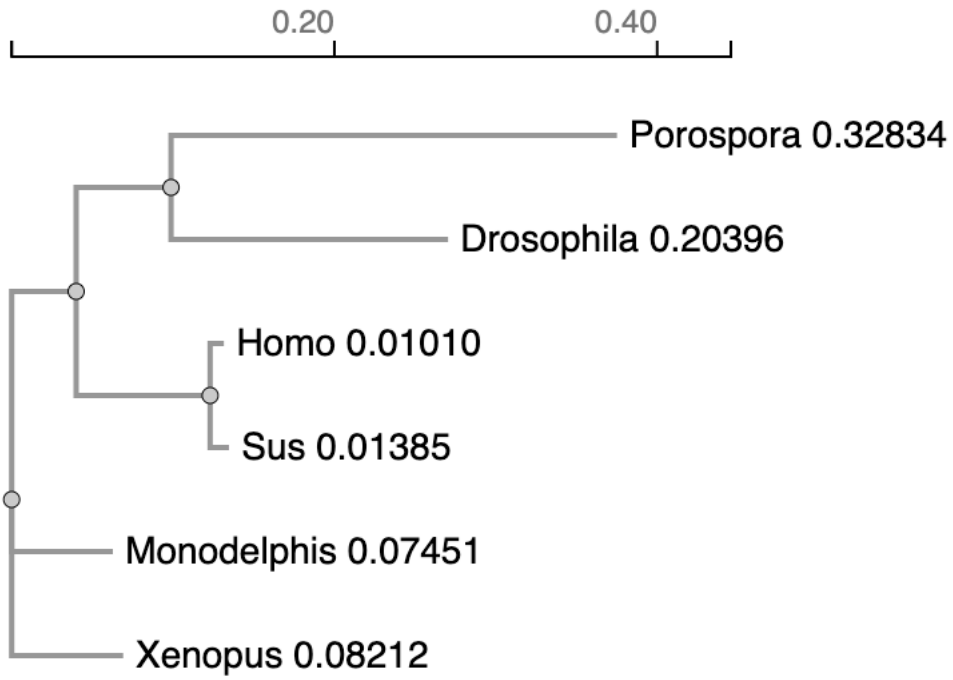
Multiple Sequence Alignment (MSA)

Job ID: muscle-I20260228-222329-0639-21984930-p1m

```
CLUSTAL multiple sequence alignment by MUSCLE (3.8)


Porospora       -------------------------------------------MSAEDESSPVILNVFHT
Drosophila      -------------------------------------------MAAIKDS--LLAQVA--
Homo            -------------------------------------------MATLKEK--LIAPVA--
Sus             MRGGPRFLLLPCRSCARADARCRLCLLRATLTAVSRTFTPLRTMATLKEK--LIAPVA--
Monodelphis     -------------------------------------------MGTVKDQ--LILNVL--
Xenopus         -------------------------------------------MATVKDK--LIHNVV--
                                                           *.: .:.  ::  *

Porospora       HRKEQTFPWTITVVGAAGQVGLACALACVSQAVGNRVCMFDVVADRLHGEVKDIQQALTF
Drosophila      -EVLPS-SGHKVTIVGIGQVGMASAFSILAQNVSKEVCLIDVCADKLQGELMDLQHGSNF
Homo            -EEEATVPNNKITVVGVGQVGMACAISILGKSLADELALVDVLEDKLKGEMMDLQHGSLF
Sus             -EEETTIPNNKITVVGVGQVGMACAISILGKSLTDELALVDVLEDKLKGEMMDLQHGSLF
Monodelphis     -KEEQT-PHNKITVVGVGAVGMACAISILMKDLADELALVDVIEDKLKGEMMDLQHGSLF
Xenopus         -KEESL-PQNKVTIVGVGAVGMACAISVLQKDLADELALVDVIEDKLKGEMMDLQHGSLF
                       .    .: . * **:*.*:: : : : . :.:.**  *.*:**: *:*:.  *

Porospora       IANPPCVFAATNLKDTANSRIVVVTAGVRQQPGESRLGLLQRNVNVFKSIIPEIAKHSPD
Drosophila      LKNPQIT-ASTDFAASANSRLCIVTAGVRQKEGESRLSLVQRNTDILKNIIPKLVEYSPD
Homo            LQTPKIV-ADKDYSVTANSKIVVVTAGVRQQEGESRLNLVQRNVNVFKFIIPQIVKYSPD
Sus             LQTPKIV-ADKDYSVTANSKIVVVTAGVRQQEGESRLNLVQRNVNVFKFIIPQIVKYSPD
Monodelphis     LKTPKIV-SSKDYAVTANSKLVVITAGARQQEGESRLNLVQRNVNIFKFIIPNIVKYSPN
Xenopus         LRTPKIV-SGKDYSVTANSKLVVVTAGARQQEGESRLNLVQRNVNIFKFIIPNIVKYSPN
                : .*  . : .:   :***.: ::***.**: *****.*:***.:::* ***::.::**:

Porospora       AVLLIATNPCDIMTYVAWKLSGFCRNRVISTGCNLDTGRFRHYLANELHCSTSSIHAFIL
Drosophila      TILLMVSNPVDIMTYVAWKLSGLPKNRVIGSGTNLDSSRFRFLMSQRLGVAPTSCHGWII
Homo            CIIIVVSNPVDILTYVTWKLSGLPKHRVIGSGCNLDSARFRYLMAEKLGIHPSSCHGWIL
Sus             CIIIVVSNPVDILTYVTWKLSGLPKHRVIGSGCNLDSARFRYLMAEKLGVHPSSCHGWIL
Monodelphis     CKLLVVSNPVDILTYVAWKLSGFPKNRVIGSGCNLDSARFRYLMGEKLGIHSSSCHGWIL
Xenopus         CTLLIVSNPVDILTYVAWKISGFPKNRVIGSGCNLDSARFRYLMGQKFGIHTQSCHGWVI
                  :::.:** **:***:**:**: .:***.:* ***:.***. :.: :   . * *.:::

Porospora       GEHGDSSVPVWSSTNVGSAIVSEL-------CDLSKTSNAHKHVVASAEEIIRLKGYTNW
Drosophila      GEHGDSSVPVWSGVNIAGVRLRELNPILGTGEDPEKWNELHKQVVDSAYEVIKLKGYTSW
Homo            GEHGDSSVAVWSGVNVAGVSLQELNPEMGTDNDSENWKEVHKMVVESAYEVIKLKGYTNW
Sus             GEHGDSSVAVWSGVNVAGVSLQELNPEMGTDNDSENWKEVHKMVVESAYEVIKLKGYTNW
Monodelphis     GEHGDSSVPVWSGVNVAGVSLKSLHPALGTDSDSEQWKDVHKQVVESAYEVIKLKGYTSW
Xenopus         GEHGDSSVPVWSGVNVAGVSLKTLHPDIGSDADKENWKEVHKQVVDSAYEVIKLKGYTSW
                ********.***..*:... :  *        * .: .: ** ** ** *:*.*****.*

Porospora       GIGFSVAQLCRRILHNTEEVAAVGTFVQGLFGIEEDVVLTVPCLLGAHGVHGVVNLTFKD
Drosophila      AIGLSTASLASAILRNTSSVAAVSTSVLGEHGIDKDVFLSLPCVLNANGVTSVVKQILTP
Homo            AIGLSVADLIESMLKNLSRIHPVSTMVKGMYGIENEVFLSLPCILNARGLTSVINQKLKD
Sus             AIGLSVADLIESMLKNLSRIHPVSTMVQGMYGIENEVFLSLPCVLNARGLTSVINQKLKD
Monodelphis     AIGLSVADLAESIMKNLRRVHPISTMIKGLYGINEDVFLSVPCILGQNGISDVVKVTLTT
Xenopus         AIGLSVADLSESILKNLRRVHPISTMVKGMYGVNNDVFLSVPCVLGNLGITDVVNMTLKA
                .**:*.*.*   ::.*   : .:.* : * .*::::*.*::**:*.  *: .*::  :.

Porospora       EENEALRASAKTLAESMKGIE-Y
Drosophila      TEVEQLQKSANIMSDVQAGLK-F
Homo            DEVAQLKKSADTLWDIQKDLKDL
Sus             DEVAQLKNSADTLWGIQKDLKDL
Monodelphis     EEESRLKQSADTLWGIQKELQ-F
Xenopus         DEEDRLRKSADTLWAIQKELQ-F
                 *   *. **. :      ::
```

[Q6] Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use “simple phylogeny” online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image

of your Cladogram or tree output in your report.



A distance-based Neighbor-Joining phylogenetic tree was generated using the EBI Simple Phylogeny tool from the MUSCLE multiple sequence alignment of the selected LDH family proteins.

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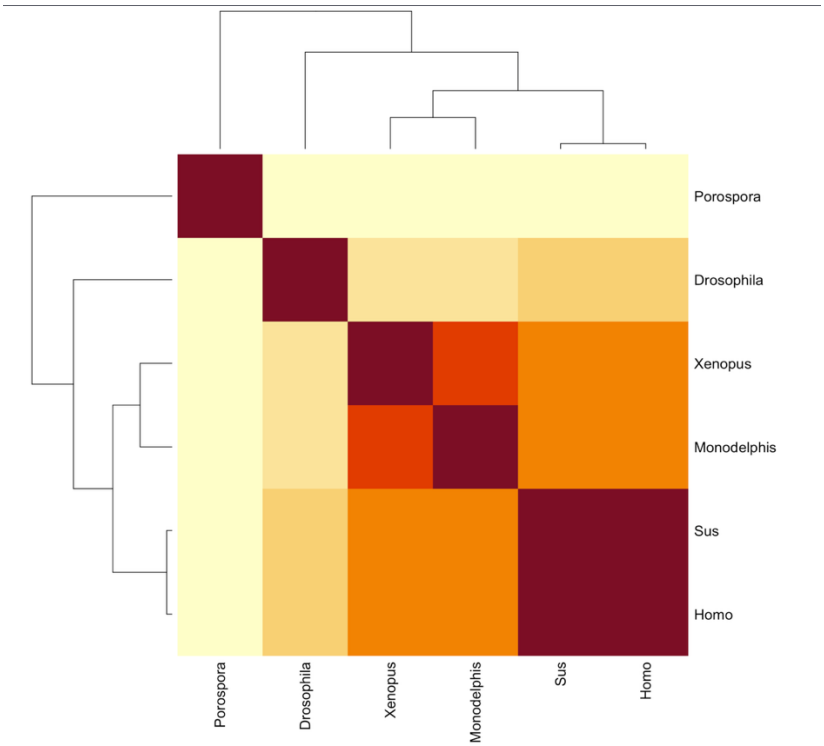
[Q7] Generate a sequence identity based **heatmap** of your aligned sequences using R. If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and "Save as" FASTA format for example). Read this FASTA format alignment into R with the help of functions in the **Bio3D package**. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.



Sequence identity heatmap generated from the MUSCLE-aligned LDH family protein sequences. Pairwise percent sequence identity values were calculated in R using the Bio3D package and visualized as a clustered heatmap. Warmer colors indicate higher sequence identity. Vertebrate sequences (Homo and Sus) show the highest identity and cluster together, whereas Porospora displays the lowest identity relative to the other taxa.

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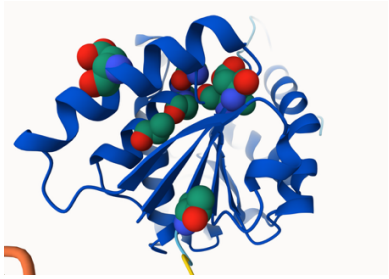
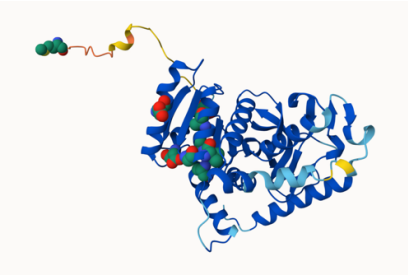
[Q8] Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences.
List the top 3 *unique* hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example, include the annotation terms PDB identifier (structureId), Method used to solve the struct

PDB ID: 7DBJ Experimental technique: X-ray diffraction Resolution: 1.551 Å Source organism: Homo sapiens Sequence identity: 50% E-value: 1.049e-90	**PDB ID**: 6CEP Experimental technique: X-ray diffraction Resolution: 2.0 Å Source organism: Sus scrofa Sequence identity: 51% E-value: 1.436e-90	**PDB ID**: 5K0Z Experimental technique: Electron microscopy Resolution: 2.8 Å Source organism: Gallus gallus Sequence identity: 52% E-value: 1.29e-89

All (3) PDB represent L-lactate dehydrogenase B chains across humans, pig, and chicken.

[Q9] Using AlphaFold notebook generate a structural model using the default parameters for your novel protein sequence.





Conserved Residues:

Met1, Val27, Gln32, Leu35, Cys37, Ala38, Ser42, Phe58, Glu70, Arg104